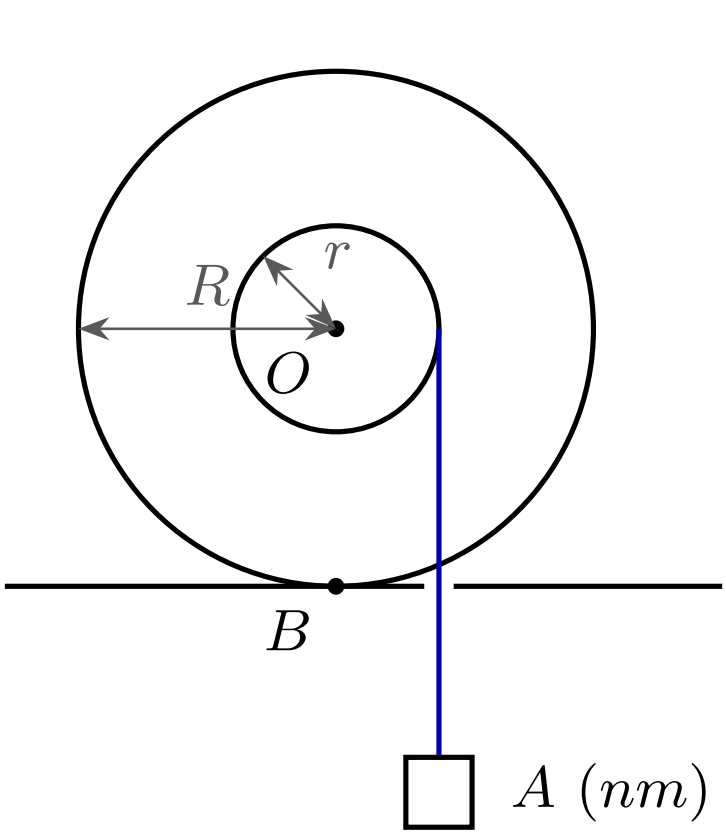
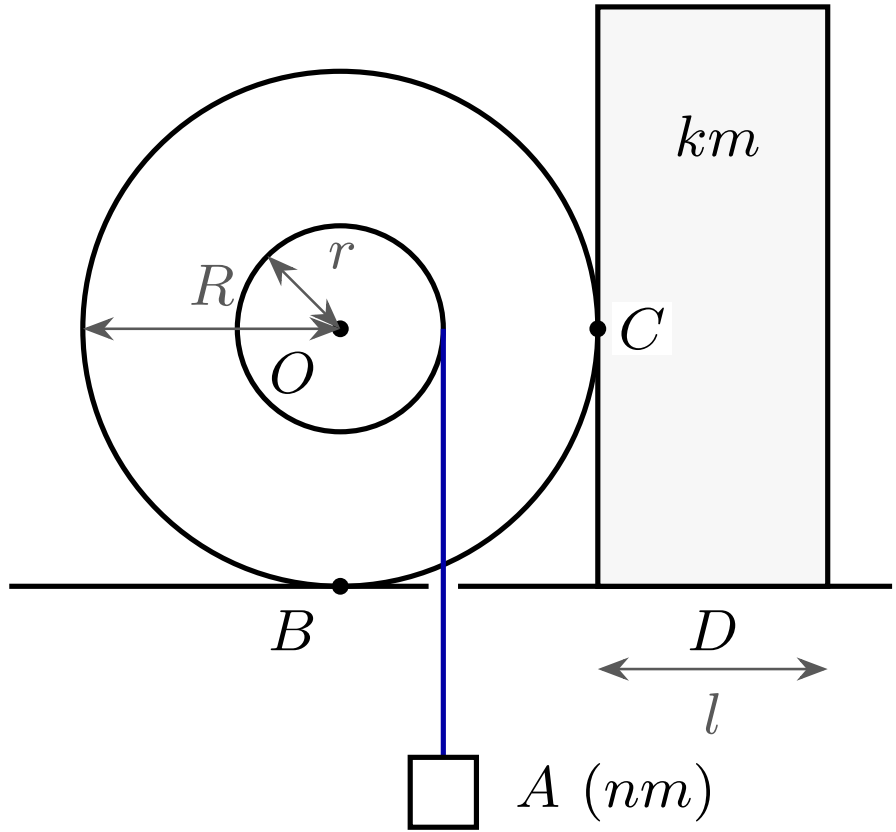
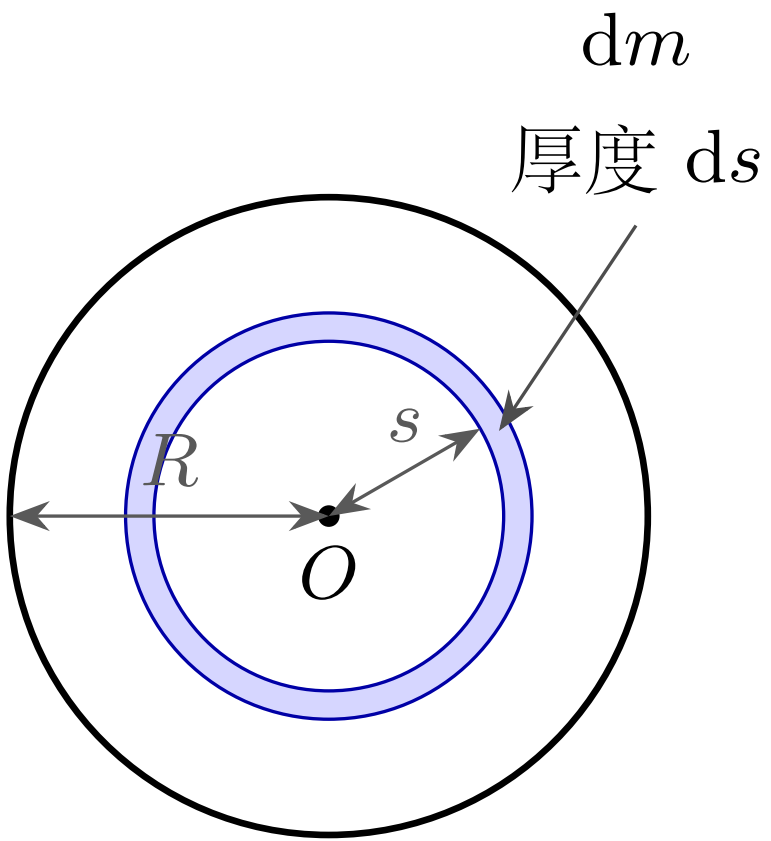


无侧面障碍

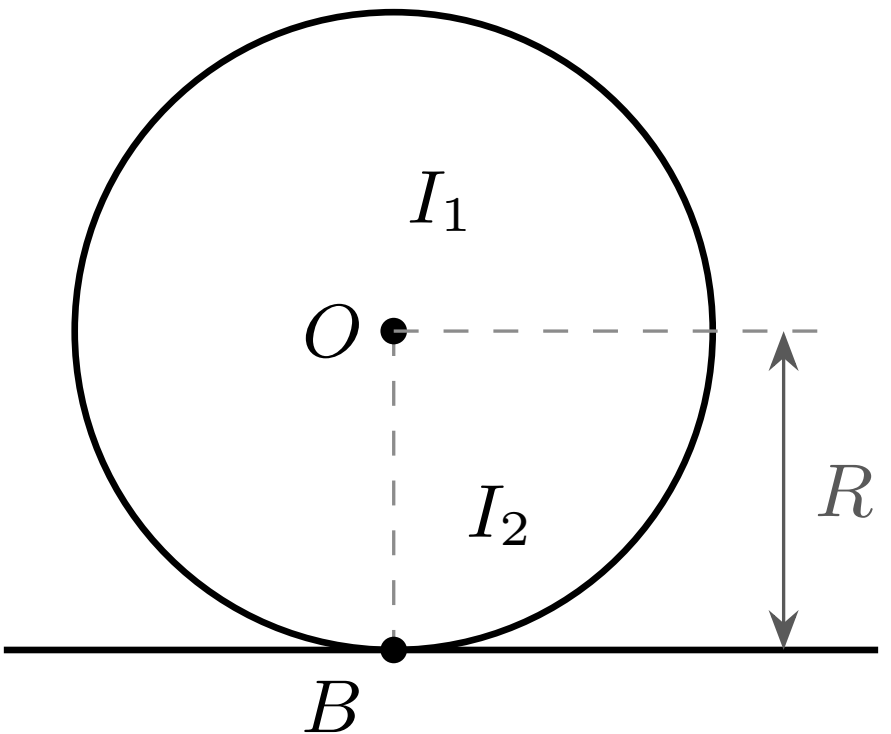


右侧长方体接触

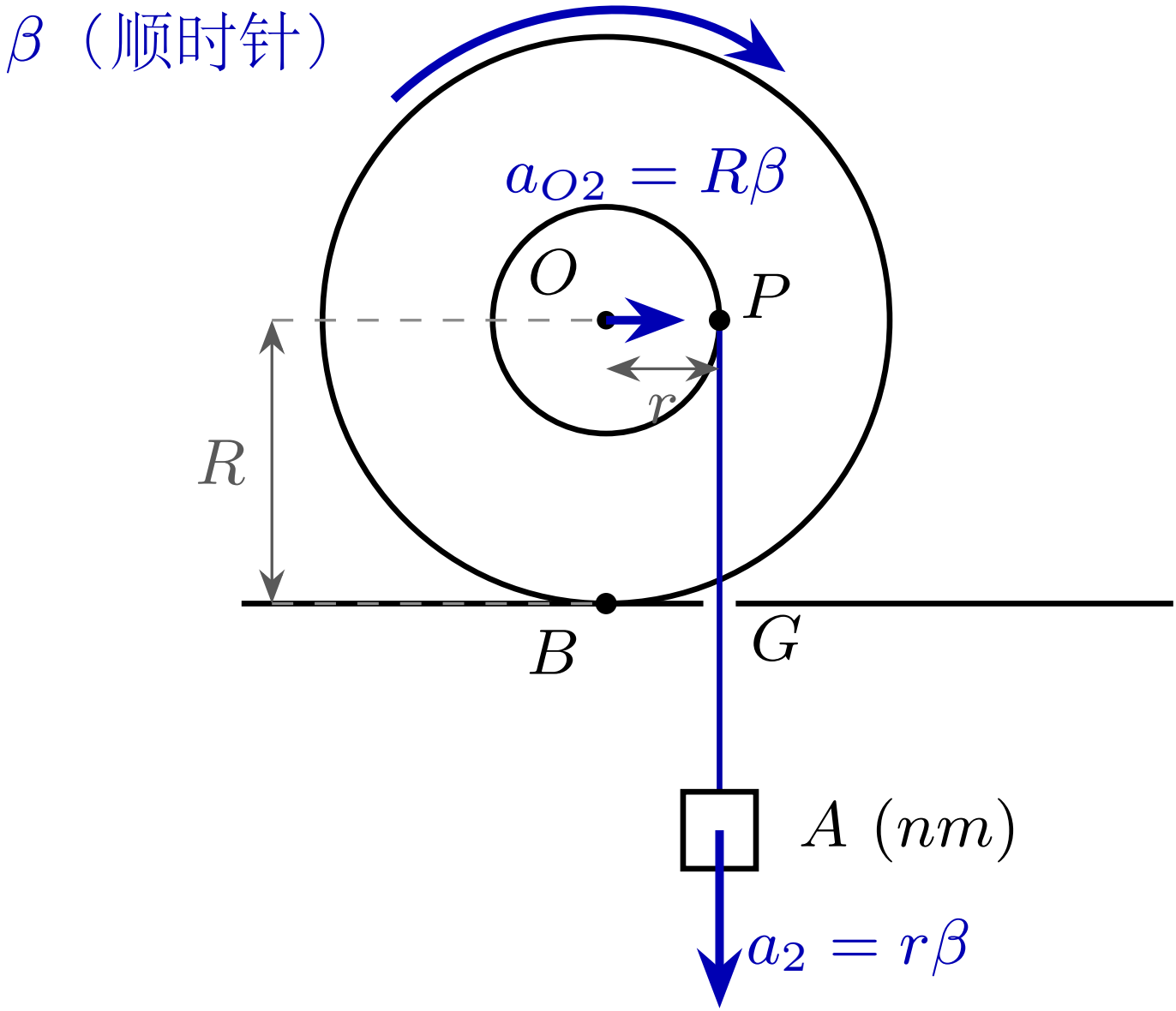
沿圆柱轴线观察；绳与内轴相切，细缝不固定绳端



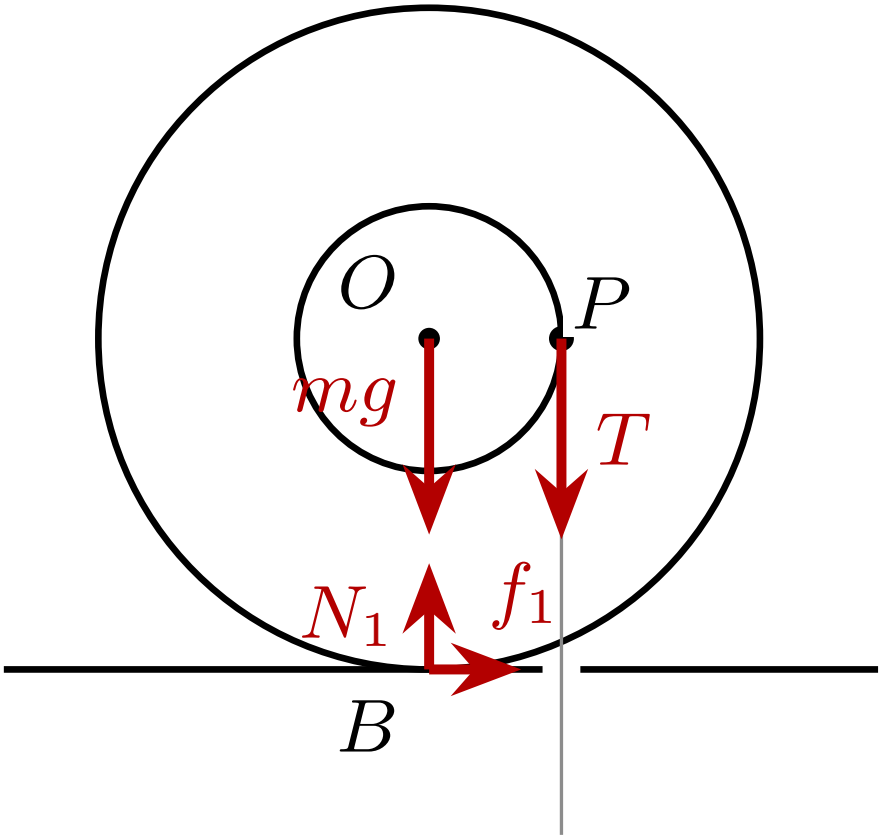
同轴薄圆筒的截面



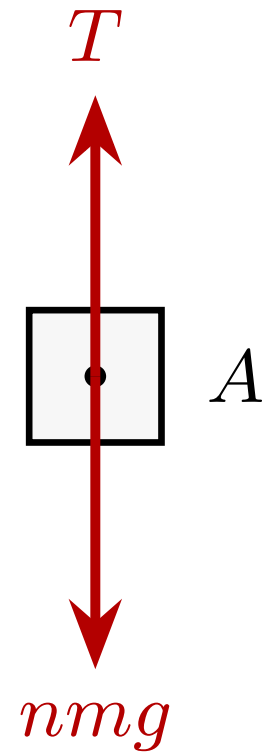
两条平行轴均垂直图面



释放瞬间；$v_O = \omega = 0$，绳初始竖直

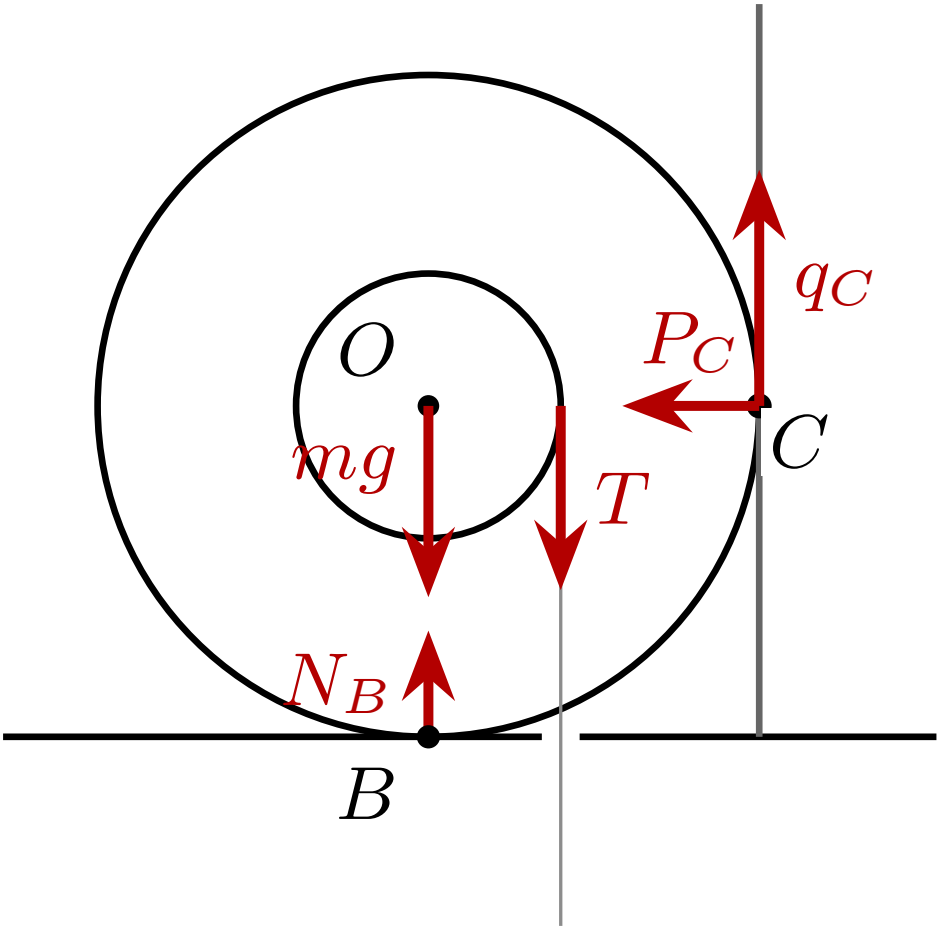


悠悠球



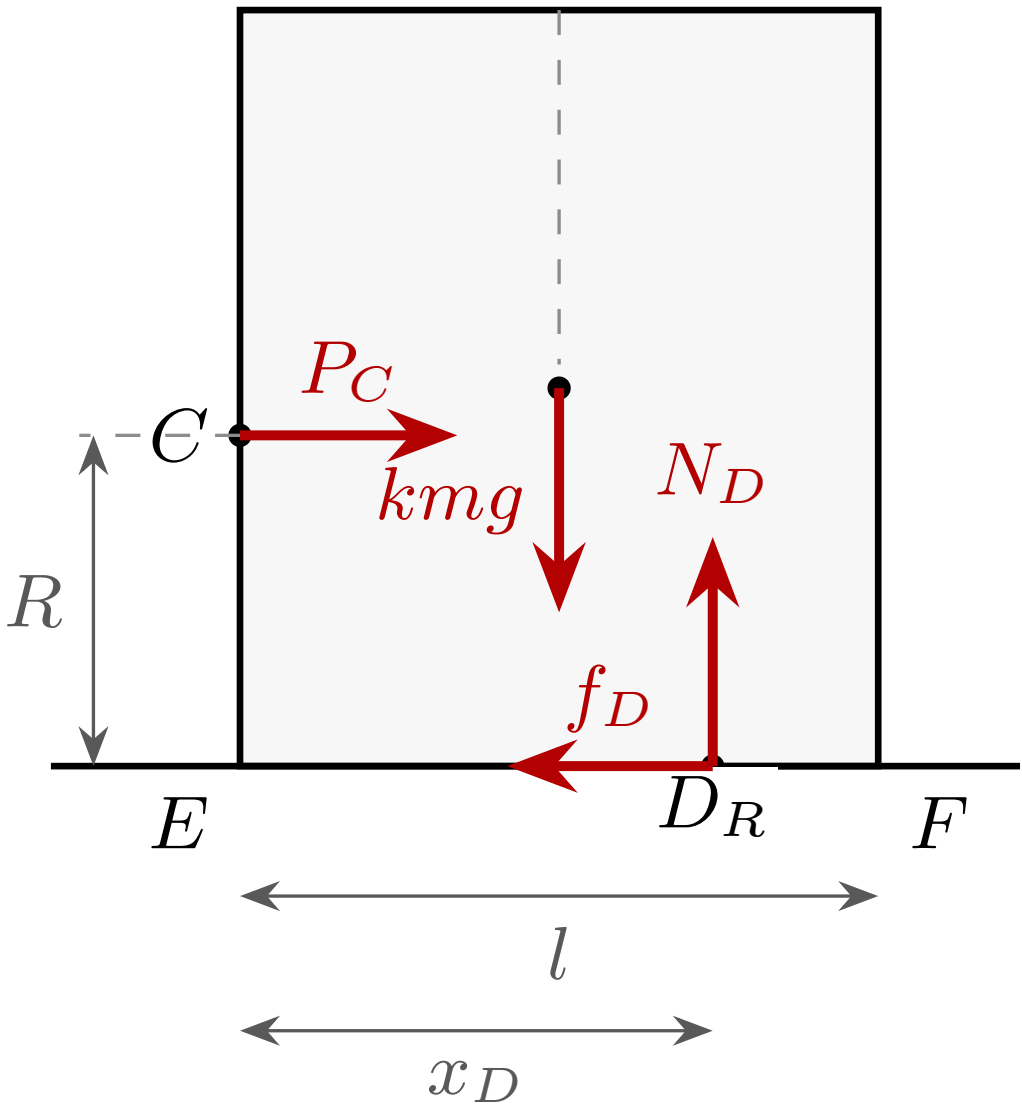
悬挂重物

长方体左侧面



B 光滑： $P_C = 0 \Rightarrow q_C = 0$

均匀长方体



$$x_D = l/2 + nr/k,\ \text{需}\ x_D \le l$$